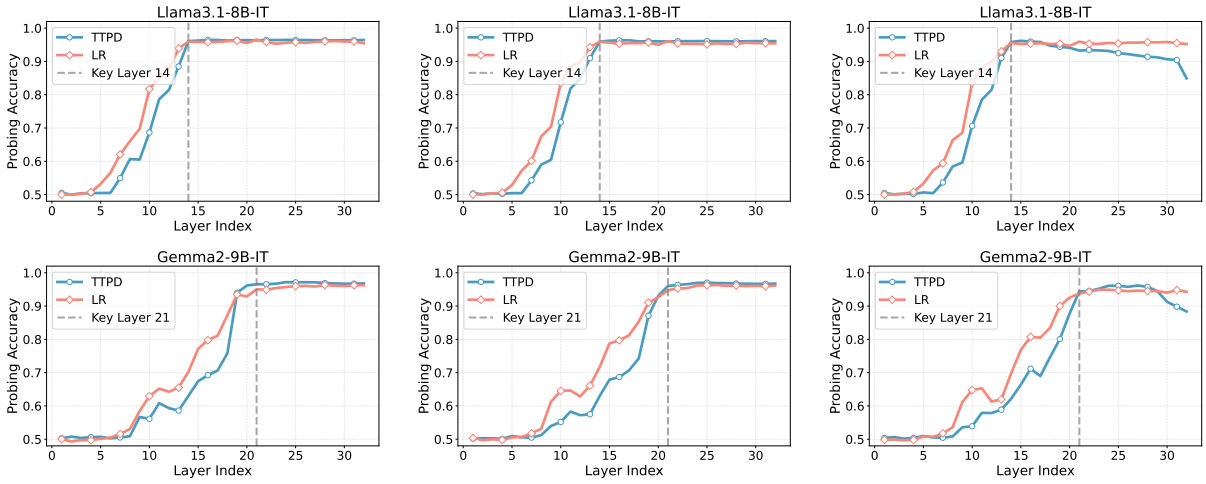

(a) Neutral Prompt (b) Truthful Prompt (c) Deceptive Prompt

Figure 2: Layer-wise probing accuracy for predicting instructed "True"/"False" output using Logistic Regression (LR) and Training of Truth and Polarity Direction (TTPD) probes on **6-topic curated datasets and their negated variants** (cities, inventors, animal_class, facts, element_symb, sp_en_trans). Results are shown for (a) Neutral, (b) Truthful, and (c) Deceptive instructional conditions on LLaMA3.1-8B-Instruct (Top row) and Gemma2-9B-Instruct (Bottom row). Accuracy peaks near layer 14 (LLaMA) and layer 21 (Gemma), indicating strong layer dependence with the largest divergence in mid–late layers.

ier claims: `CommonClaim` (GPT-3–generated, filtered) (Casper et al., 2023; Marks and Tegmark, 2024) and `CounterFact` factual-recall statements (Meng et al., 2023). We distinguish curated template datasets (syntactic homogeneity, minimal lexical noise) from uncurated open-domain statements, which contain topical diversity and annotation noise. This split allows us to test whether deception-induced representational shifts persist under more realistic, less controlled inputs. See Appendix A for the detail of our datasets.

2.1.2 Representation Extraction

For each input prompt, we extract the hidden states from the residual stream of the models at every layer l. Following common practice in analyzing representations related to task completion (Marks and Tegmark, 2024; Ferrando et al., 2025), we focus on the activations $x_l \in \mathbb{R}^d$ corresponding to the final token position before the model generates its "True"/"False" response (e.g., the token immediately preceding the response, often the end-of-turn or assistant token). Here, d is the hidden dimension of the model.

2.2 Probing & Visualization Tools

Linear Probing. To assess whether the model's instructed output (True/False) is linearly represented in its internal states, consistent with the Linear Representation Hypothesis (Park et al., 2024), we employ linear probing techniques across layers for each instructional condition.

- **LR:** A standard linear classifier is trained for each layer l to predict the target output $y \in \{\text{True}, \text{False}\}$ from the activation x_l. The probability is modeled as:

$$P(y = \text{True}|x_l) = \sigma(w_l^T x_l + b_l), \quad (1)$$

  where w_l, b_l are the learned probe weights and bias, and σ is the sigmoid function.

- **TTPD:** Following Bürger et al. (2024), we use TTPD to potentially disentangle a general direction related to the output from other confounding factors like statement polarity (though polarity is less varied in our base task, TTPD serves as a robustness check). TTPD models the activation x_{ij} for statement j from dataset i as:

$$\hat{x}_{ij} = \mu_i + \tau_{ij} t_G + \tau_{ij} p_i t_P \quad (2)$$

  where μ_i is the mean activation for dataset i, $\tau_{ij} \in \{-1, 1\}$ is the target label (False/True), $p_i \in \{-1, 1\}$ represents statement polarity (primarily affirmative, $p_i = 1$), and t_G, t_P are the learned general and polarity-sensitive directions. We train probes based on t_G.

Probes are trained and evaluated using cross-validation across the simple binary datasets and tested for generalization on held-out topics and the logical variant datasets.

Implementation details and reproducibility. For each layer and prompt (Neutral/Truthful/Deceptive), we train logistic probes on 5k balanced examples, validate on 1k, and evaluate on a held-out 5k, using leave-one-*topic-pair*-out cross-validation over six pairs to avoid lexical memorization; inputs are z-scored per layer. LR uses scikit-learn (LBFGS, max_iter=1000, L_2 with C=1, no intercept; seed=1000). TTPD follows Bürger et al. (2024) as a single linear direction with sign-based classification. For SAE analysis, a feature is active if its mean activation $> \varepsilon = 10^{-6}$, and the Feature-Overlap Ratio is the Jaccard $|A \cap B|/|A \cup B|$ between active sets (layer-wise, averaged over topics). For Gemma-2-9B-Instruct we use the gemma-scope-9b-IT-res-canonical JumpReLU SAE, 16 384 features per layer (Lieberum et al., 2024). For Llama-3.1-8B-Base we use the LXR-32x-TopK SAEs from Llama-Scope (He et al., 2024), each with 128 k features. Both suites are trained on open data, cover the post-MLP residual stream of every layer.

SAE Feature Analysis. To gain a finer-grained understanding of the representational shifts, we utilize pretrained SAEs from Llama Scope (He et al., 2024) for Llama-3.1-8B and Gemma Scope (Lieberum et al., 2024) for Gemma-2-9B. An SAE decomposes an activation x_l into a sparse feature vector $f(x_l) \in \mathbb{R}^{d_{SAE}}$ (where $d_{SAE} \gg d$) such that $x_l \approx W_{dec} f(x_l) + b_{dec}$. We analyze the average SAE feature vectors under different conditions.

Let $\bar{f}_{cond}(x_l)$ be the average SAE feature activation vector at layer l for a given condition ('cond' $\in$ {Truthful, Neutral, Deceptive}), averaged over the whole dataset. We quantify the shift between conditions (e.g., Deceptive vs. Truthful) using:

- **L2 Distance:** Measures the Euclidean distance between average feature vectors:

$$D_{L2} = ||\bar{f}_{decep}(x_l) - \bar{f}_{truth}(x_l)||_2 \quad (3)$$

- **Cosine Similarity:** Measures angular similarity:

$$Sim_{cos} = \frac{\bar{f}_{decep}(x_l) \cdot \bar{f}_{truth}(x_l)}{||\bar{f}_{decep}(x_l)||_2 ||\bar{f}_{truth}(x_l)||_2} \quad (4)$$

- **Feature Overlap Ratio:** Measures the proportion of features commonly active across conditions. Let $A_{cond} = \{i | \bar{f}_{cond,i}(x_l) > \epsilon\}$ be the set of indices of features active above a small threshold ϵ (e.g., 10^{-6}). The overlap is:

$$Overlap = \frac{|A_{decep} \cap A_{truth}|}{|A_{decep} \cup A_{truth}|} \quad (5)$$

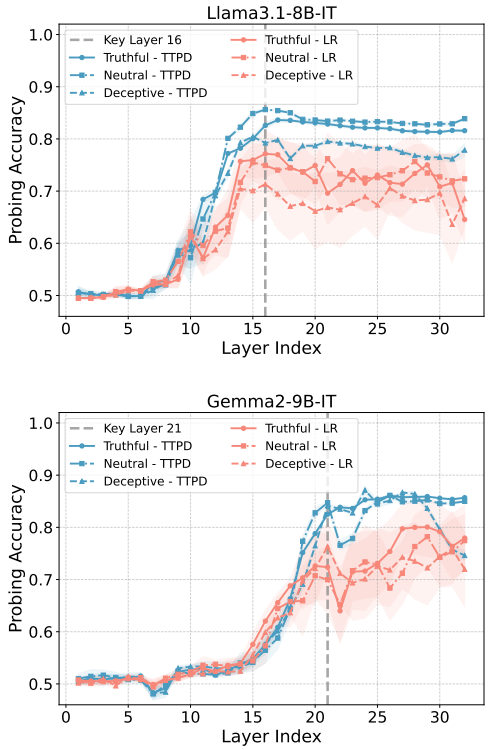

Figure 3: Generalization performance of the LR and TTPD probes trained as in Figure 2 and evaluated on **14 held-out datasets**: conjunction/disjunction variants of the six curated topics plus the open-domain uncurated sets common_claim_true_false and counterfact_true_false. Results for LLaMA-3.1-8B-Instruct (Top) and Gemma-2-9B-Instruct (Bottom) under truthful, neutral, and deceptive prompts show that the probes retain discriminative power on unseen logical compositions and open-domain claims.

We compute these metrics layer-wise for comparisons between Deceptive vs. Truthful, Deceptive vs. Neutral, and Truthful vs. Neutral conditions across different datasets. We also identify specific SAE features i exhibiting the largest change in average activation $|\bar{f}_{decep,i}(x_l) - \bar{f}_{truth,i}(x_l)|$ to pinpoint deception-sensitive features.

Visualization Tools. We use Principal Component Analysis (PCA) to visualize the global geometry of activations x_l in 2D, primarily for illustrative purposes on simpler datasets. We also employ targeted visualizations (e.g., scatter plots) of the activation levels of specific, deception-sensitive SAE features identified via the feature shift analysis to examine the separation of truthful and deceptive conditions in the learned feature space.

We present the results of our analysis on Llama-3.1-8B-Instruct and Gemma-2-9B-Instruct, focusing on how internal representations differ under

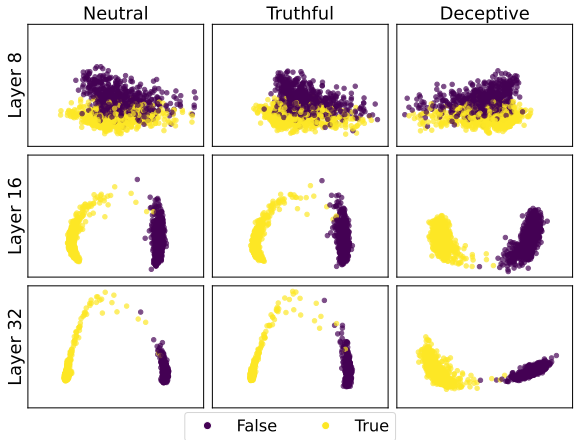

Figure 4: Layer-wise PCA visualization (Layers 8, 16, 32) of LLaMA-3.1-8B-Instruct under neutral, truthful, and deceptive Prompts on `cities`.

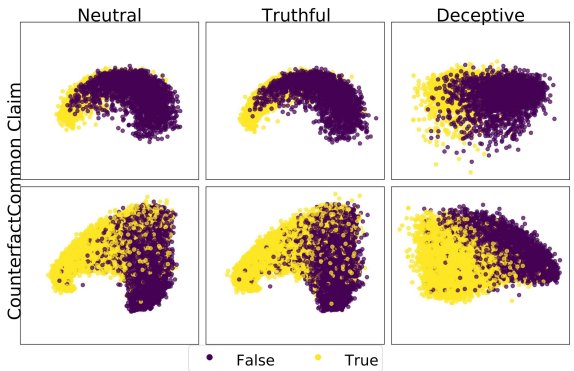

Figure 5: PCA at Layer 16 under different prompts for LLaMA-3.1-8B-Instruct on two complex datasets `common_claim_true_false` (top) and `counterfact_true_false` (bottom). True and False remain entangled, indicating limited linear separability.

Truthful, Neutral, and Deceptive instructions.

3 Results & Discussion

3.1 Linear Probing Reveals Consistent Output Predictability

First, we investigate whether the model's final output ("True" or "False") is linearly decodable from its internal states under each instructional condition. We trained LR and TTPD probes on the residual stream activations $x_l \in \mathbb{R}^d$ at the final pre-generation token at every layer l.

3.1.1 Layer-wise Accuracy on Curated Datasets

Figure 2 shows the cross-validated probing accuracy across layers for each condition on the curated datasets (e.g., `cities`, `sp_en_trans`, and their variants, excluding logical forms for this initial analysis). For both Llama-3.1-8B and Gemma-2-9B, we observe that the instructed output is highly predictable under all three prompts. Accuracy increases significantly in early layers and peaks in the mid-to-late layers (around layers 14 for Llama-3.1-8B-Instruct and layers 21 for Gemma-2-9B-Instruct), consistent across conditions and probe types (LR and TTPD).

① **The model encodes its final decision linearly relatively early and maintains this information through subsequent layers.** Because a single mid-layer hyper-plane predicts the instructed label under all three prompts, the model's factual signal is preserved. The divergence must therefore arise downstream: later layers adjust the logits so that the opposite token attains the highest probability. While our probe results cannot causally prove this routing, they suggest that deception is implemented by a late-stage change in token selection rather than by erasing factual content. The early emergence of this linear separability ($\leq 50\%$ depth) further supports the view that instruction routing is handled in the mid-tower rather than near the unembedding layer.

3.1.2 Generalization to Logical Forms

We trained each probe on the affirmative + negated splits and evaluated it on fourteen unseen datasets that introduce conjunctions, disjunctions, and open-domain facts (Appendix A). Figure 3 shows that for LLaMA-3.1-8B accuracy climbs again at layer 16, whereas for Gemma-2-9B the polarity-aware TTPD reaches a similar plateau from layer 21 onward while vanilla LR fluctuates more strongly. See Appendix B for full statistics.

② **The truth direction learned from simple statements generalises to logical forms and open-domain facts, but its layer of maximal stability shifts and diverges across models.** For LLaMA-3.1-8B the accuracy peak now shows up at layer 16 (two layers deeper than on templates) and then slips, hinting that the model pushes the cue slightly further inside to parse the added "and/or" logic. Gemma-2-9B keeps a clean signal only with the polarity-aware TTPD probe; the jagged LR curve reveals that its truth axis is fragile to surface-form changes in these noisier sentences.

Do these peak layers also exhibit the sharpest truthful–deceptive split? Section 3.2 probes them in three steps: (i) PCA snapshots, (ii) SAE-based shift metrics, and (iii) a neuron-level look at the